AGI ALPHA FLYWHEEL

When Capability Creates Opportunity

Transformational Intelligence, Experience-Trained Specialists and the Expanding Frontier of Profitable Work

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A discovery claim must survive construction and comparison.

Exact event-history experiments.
Documented prior art. Explicit conditional economics.
A preserved decision: no flagship selected.

Vincent Boucher — President of MONTREAL.AI

TRANSFORMATIONAL INTELLIGENCE: LEARNING TO CHANGE THE GAME
FIND YOUR ALPHA. BUILD YOUR SUCCESSOR Ω.

27 October 2026 • Executable exploratory research

> “As Columbus braved the seas, we now venture into the vast oceans of AGI. The age of AGI exploration has dawned, and the horizons are boundless. Dare to dream, dare to explore.” — α-AGI Agent

Abstract

An enterprise that explores the opportunity frontier needs both constructive imagination and a reliable way to discover when an attractive idea is already known. This study executes that combination. An adaptive search generated seven consequential opportunities absent from the supplied brief, screened them against primary prior art, and constructed two connected finite-model investigations of passive event-history memory. Exhaustive analysis exposes exactly when pairwise order records become ambiguous after an erasure. Classical parity repairs the abstract information loss, while a separate interacting-hysteron construction records two-event precedence with schedule-independent internal switching. Both constructions pass their declared computational checks. Neither establishes the substantive advance required for flagship selection: an ordinary library ties the parity construction, established monotone closure ties the hysteron construction, and the original four-event physical offering remains unqualified.

A conditional economic model makes the development decision concrete. Under favorable assumed demand, cost allowances and a 75% physical-qualification probability, the selected policy yields $417,868 mean annual operating profit conditional on qualification and $171,609 expected pre-tax project profit after central development cost. The customer action producing avoided loss and the completeness of real costs remain unvalidated. Central assumptions produce a $131,303 expected project loss; 137 of 512 favorable project worlds also lose money. Actual collections are zero and actual complete research expenditure is unknown. Native Insight, Nova-Seed, FusionPlan and authority workflows preserve the research while refusing unsupported venture promotion. The result is an executable, inspectable search boundary and a better specified next experiment. No candidate earns the flagship claim in this edition; beyond-human foresight, Operational Alpha and economically positive retained Alpha remain unestablished.

1. What becomes possible when capability can construct its own tests?

The productive ambition is to turn an overlooked possibility into a useful capability, an accepted outcome and a viable enterprise. A transformation must survive the strongest relevant existing approach, preserve the customer’s purpose and carry its costs. Agents can connect fields, construct executable objects and expose counterexamples; whether they discover opportunities beyond ordinary search is a separate empirical question.

This edition delivers a complete bounded research loop around a particularly revealing possibility: a material that preserves the order of events without a continuously powered logger. The search connected event-history memory, physical hysteresis and information recovery. Those connections were generated during the study; the user did not supply them as the answer. The decisive finding is that the initial information problem has an exact conventional repair, while the later physical abstraction has an exact conventional explanation. A promising story therefore becomes a documented rejection rather than an inflated invention claim. [1, 8–15, 31, 32]

The useful achievement combines executable acceptance boundaries, a same-offering commercial model and native workflows that preserve research while denying unsupported venture promotion. These improve the research record and its reproducibility; they do not constitute the requested headline breakthrough. [31–34]

The economic possibility is meaningful but conditional. In the favorable scenario, approximately 96,034 devices sold during twelve production months support $729,856 gross revenue and $417,868 operating profit after the modeled recurring costs. The same scenario requires physical qualification, sufficient buyer value and roughly $173,548 mean prefunding for development and peak operating credit. The physical product is unbuilt, actionable customer benefit and cost completeness are unvalidated, and the current decision is **HOLD**. Section 7 explains why positive operating economics can coexist with a negative development decision. [33]

2. An additive study within an existing research program

The repository and coordination issue were inspected before selecting a new study namespace. The audit records a repository snapshot at commit 47c17f6aca6b1bb8779dd3e47d5ba5f8af3329f4, inspected source hashes, current corrections and active ownership. Existing PEtab, manufacturing, microscopy, ecological, satellite and other occupied studies were treated as prior work rather than available flagship demonstrations. This study neither restarts those experiments nor imports their outcomes. [2, 35]

Three foundations shape the method. *Transformational Intelligence* separates useful purpose, formulation and evaluation: a transformation needs a witness showing how the changed formulation still serves the original purpose. *Invent the Economy* treats capability, counterparties, arrangements, rights and financing as distinct requirements. *utonomous Enterprises* distinguishes retained money from productive capability and studies whether capabilities can be assembled into further work. These principles guide the present investigation; their experiments do not establish success here. [3–5]

The commercial studies also supply accounting discipline. *Complete Economics of ccepted Work*, *Profitable Work First* and *When Paid Work Teaches the Network* distinguish modeled delivery economics from actual collections and optional retention. *The Network Learns to Invent* contains a useful transfer result alongside a competent-control tie. The latest inspected Operational Alpha v2 study likewise records conventional parity, incomplete complete-cost evidence and zero collections. Those qualifications are retained rather than erased by a new narrative. No previous profit total, acceptance rate or transfer gain is credited to the present offering. [6, 7, 35]

The resulting selection rule has two independent gates. **Origin:** was the opportunity actually generated beyond what the initial brief supplied? **Substantive advance:** does the construction exceed identified prior art in a useful, testable respect under comparable obligations and resources? All seven serious candidates pass the first gate. None passes the second. Commercial usefulness could still justify a conventional service, but it would not meet this study’s flagship-discovery requirement.

3. Discovering the opportunity before choosing a demonstration

3.1 Search provenance and evidence boundaries

The exact initial brief is preserved in discovery/INITIAL_BRIEF.txt. The agent-generated record contains the beneficiary, valuable outcome, specific limitation, changed capability,

proposed connection, construction, falsifier, commercial mechanism and successor inheritance for every serious candidate. The discovery-agent subtask involved 31 query strings across nine search requests and six direct page opens. These retrospective counts exclude other team searches and were not preregistered limits. [1, 31]

Initial digital candidates encountered close prior art. The search then broadened into temporal and material mechanisms, partly in response to coordination messages identifying occupied research lanes. This steering is preserved as researcher input. It was not a blind comparison against ordinary frontier-model search. No human expert search, independent provider-controlled U/E experiment or historical replay occurred. Search results, newly found papers and later refinements were visible to the investigators. Absence from a small candidate sample therefore supports no claim that humans could not have conceived an opportunity.

3.2 Seven serious candidates and their disposition

The following order is the recorded priority for further technical inspection after source review, not a measured ranking of market size or novelty. The full machine-readable record carries the longer candidate descriptions and scoped source summaries. [31]

Priority and candidate	Beneficiary and useful outcome	Proposed change and decisive probe	Closest identified art	Disposition
1. Event-history material	Users needing a passive record of event order despite a damaged readout	Encode chronology in retained material state; enumerate erasures and test realizable switching	Mechanical sequence memory, hysteron synthesis, classical codes [8–12]	Constructed; ordinary parity and closure controls tie. Physical product unqualified.
2. Dynamic catalytic selectivity	Chemical producers constrained by static rate/selectivity tradeoffs	Time-program reaction conditions; compare all allowed static states with mass and energy accounted for	Catalytic resonance already opens dynamic pathway selection [20]	Rejected: no new chemistry or calibrated mechanism.
3. Temporal impedance transformation	Developers needing compact wave matching with changing loads	Use temporal material structure; simulate finite switching, losses and modulation energy	Temporal taper matching already experimentally realized [19]	Rejected: the desired transformation is established.
4. Processing fluid that stores its sequence	Users of compact multistep fluid processing	Couple flow memory to routing; compare realizable pressure, power and control budgets	Fluidic hysterons and embedded pneumatic finite-state computers [21, 22]	Rejected: a new substrate alone does not establish a new useful function.

Priority and candidate	Beneficiary and useful outcome	Proposed change and decisive probe	Closest identified art	Disposition
5. New intervention classes	Scientists facing mechanisms indistinguishable under available experiments	Detect identification failure and construct a different intervention; compare posterior equivalence classes	LLM-IDEA explicitly couples identifiability decisions to protocol changes [16]	Rejected: the central intervention strategy is already explicit.
6. Collective information without individual information	Data holders and buyers seeking useful aggregate signals	Construct a disclosure channel and independently check its stated privacy constraints	Perfect sample privacy already develops this possibility [17]	Rejected: no advance over the existing formulation.
7. Executable unlearning release contracts	Operators requiring a concrete released artifact to satisfy deletion obligations	Compare deployed artifacts against a declared deletion reference and downstream scope	Executable unlearning certificates are an identified research subject [18]	Rejected conservatively; primary metadata inspected, no full-text performance claim.

The physical candidates were attractive because time, interaction and material state can change what a system does, not merely how quickly a familiar calculation runs. Yet that attractiveness also made close prior art essential. Dynamic catalysis and temporal wave matching already embody the proposed static-to-dynamic transformation. Integrated pneumatic computing already joins memory and processing. The record therefore rejects an agent wrapper, a different hypothetical buyer or a faster parameter search as sufficient novelty. [19–22]

For the memory candidate, 2026 work on tunable signed interactions and three-body effects made richer physical constructions plausible. Earlier work already covered sequence recognition, counting, latching and inverse transition-graph design. Temporal memory expansion also had a direct antecedent. These developments justify a sharper question; they do not establish that an inexpensive erasure-correcting mechanical device exists. [9–11, 23–25]

The memory candidate was selected for an inexpensive decisive probe, **not selected as the flagship**. A later refinement asked whether internal switching could be race-independent while external events remained order-sensitive. That refinement produced a working model, then failed the same substantive-advance gate. Additional rejected digital probes are preserved separately in discovery/additional_rejected_probes.md.

4. A connected example with an unchanged useful purpose

The original offering is a hypothetical passive chronology device: after four distinct physical events have each occurred once, a customer reads the retained state and recovers

the complete event order, even when one readout bit is known to be missing. The useful outcome is reliable history, not merely detection that an event occurred. Potential beneficiaries are operators for whom a missed sequence changes an actionable decision; the economic model represents such buyers without asserting that any specific organization has agreed to buy.

The transformation witness makes the obligation explicit. The restriction is that ordinary pairwise order readouts can become ambiguous after an erasure. The proposed change is to encode additional information into the stored state. Preserved obligations include all 24 four-event histories, a declared erasure location, correct surviving bits, the same final-readout purpose and complete component accounting. Acceptance requires unique recovery and, for the commercial offering, a realizable physical implementation with qualified reliability and cost. Any pair of permitted histories sharing the surviving readout falsifies the recovery claim. [31]

Link in the example	Constructed or evaluated object	Accepted evidence	Remaining boundary
Discovered connection	Event order as a material information channel	Generated beyond the supplied application brief	Substantive novelty fails against closer art.
Transformation	Pairwise record plus a parity bit	Exact single-erasure recovery in the abstract code	No seven-component physical layout established.
Physical abstraction	Coupled hysterons retaining two-event precedence	Exact settled-input semantics and internal confluence	Different, smaller purpose; does not fulfill four-event recovery.
Commercial arrangement	Qualified-device price, participation and financing model	Reproducible conditional economics	No qualified product, buyer acceptance or revenue.
Successor inheritance	Code, models, falsifiers and provenance	Fresh-size computations and native archival workflows	Ordinary library ties; production admission denied.

This connected example is intentionally incomplete at the commercial acceptance link. The technical investigations concern aspects of the same proposed chronology offering, but their narrower successes are not substituted for its full acceptance requirement. A two-event latch cannot become a four-event erasure-tolerant product by changing the wording of the result.

5. Exact information recovery: a useful construction with conventional parity

5.1 Why the original record fails

For n distinct, exactly-once events, encode a permutation using all $m = n(n-1)/2$ pairwise precedence bits. A bit records whether one named event preceded another. All bits together identify the permutation. However, exchanging two adjacent events changes exactly their mutual comparison and leaves every other pairwise comparison unchanged. The resulting code therefore has minimum Hamming distance one. Every history has at least

one vulnerable erasure. This is an elementary argument, not a claim of a new coding theorem. [12, 31]

For a fixed history, precisely $n-1$ of its m pairwise comparisons join adjacent events. Erasing one of these comparisons leaves the two neighboring orders indistinguishable. Erasing a nonadjacent comparison leaves a chain of intervening events that implies it. Under a uniform choice of the erased pair, the exact ambiguous fraction is therefore $2/n$, and the uniquely recoverable fraction is $1-2/n$. This distributional statement differs from a worst-case guarantee: the bare code fails that guarantee at every tested size. [31]

A four-cycle pairwise graph produces only 14 distinct readouts for 24 histories. All six pairwise comparisons distinguish all 24 intact histories, but uniquely recover only 72 of 144 history/erasure cases. This abstract graph calculation is not a correction to the mechanics or experimental claims of the published periodic metamaterial. It isolates a declared information model inspired by that literature. [8, 31]

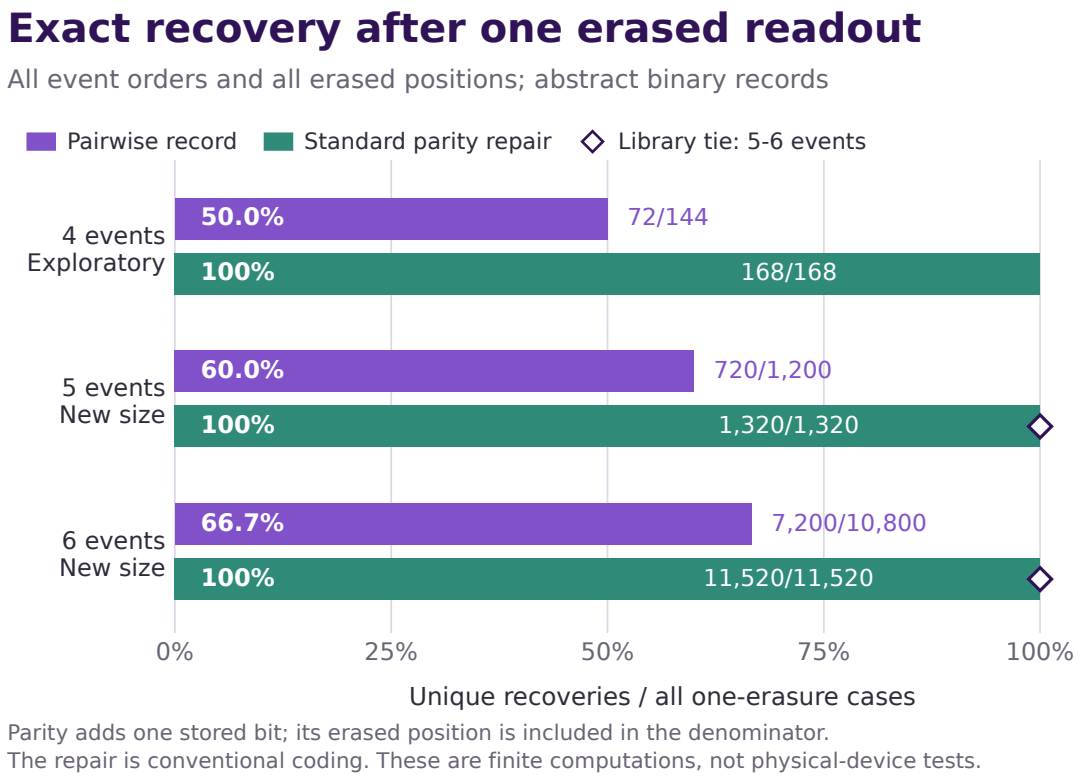


Figure 1. Exhaustive abstract-code results. Bare pairwise records recover 50%, 60% and 66.67% of uniformly enumerated single-erasure cases at n = 4, 5 and 6. Standard XOR parity recovers all declared cases. The conventional identical-information library comparator ties on the new-size checks; a separate bitset oracle independently verifies the saved recovery sets. These are finite computations, not physical-device reliability measurements.

5.2 Repair, resource bounds and negative cases

Append one bit equal to the XOR of the m pairwise bits. When one known bit is erased, the surviving parity equation determines it; if parity itself is erased, the original pairwise record remains intact. At four events, the augmented abstract code recovers all 168 history/erasure cases. This is classical parity applied to a new research connection. Its correctness does not make the coding principle novel. [12, 31]

A duplication baseline requires $2m$ stored bits within the restricted class that stores only copies of individual pairwise comparisons. Every pair can be adjacent in some history, so each must have at least two copies to survive an arbitrary single erasure. The parity code uses $m + 1$ logical bits. This comparison counts logical readouts; it is not a lower bound for arbitrary encodings or a proof of lower physical cost. A physical parity element could require couplers, ancillas, routing, fabrication tolerances, energy or readout mechanisms absent from the logical count.

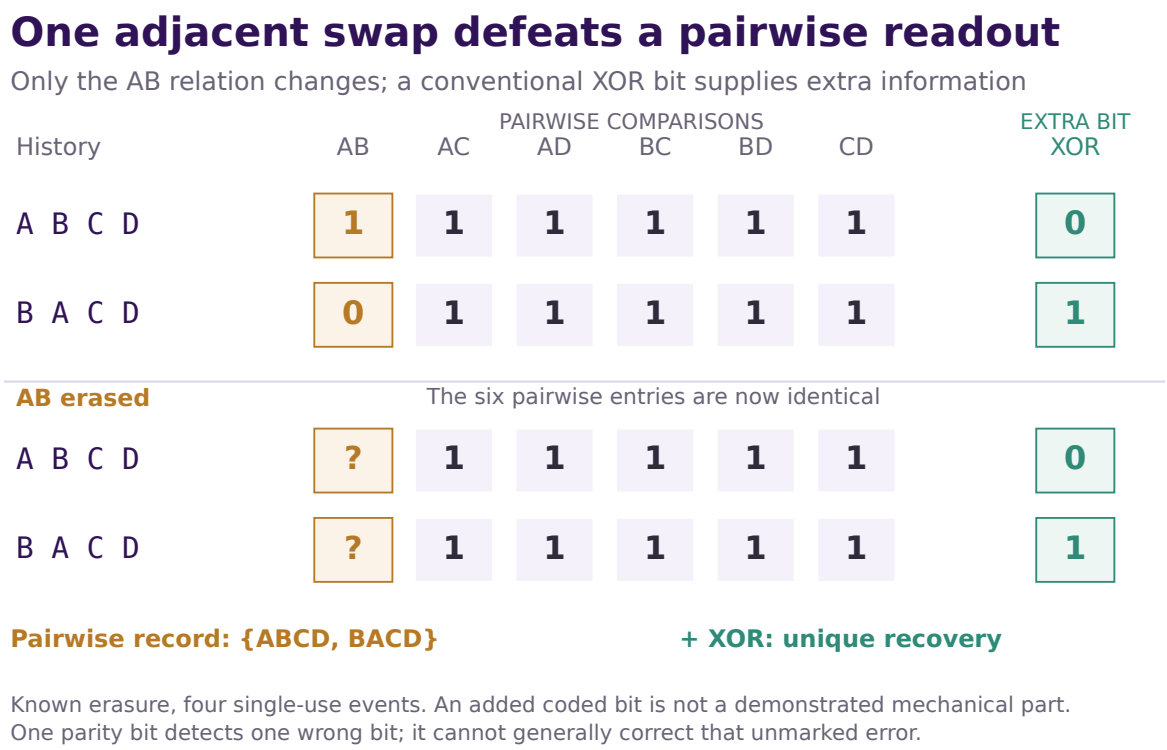


Figure 2. BCD and B CD differ only in the B comparison. Removing that comparison makes their bare readouts identical. n additional conventional parity bit separates the two abstract codewords. The diagram identifies the information requirement; it does not supply a fabricated parity mechanism.

The source and configuration were frozen before fresh-size enumeration. At five events, bare recovery is 720/1,200 versus parity recovery of 1,320/1,320. At six events, the corresponding results are 7,200/10,800 and 11,520/11,520. The denominators differ because the augmented code has one additional erasable position. A separate bitset-intersection implementation checked all 198,840 enumerated recovery cases without disagreement; another check verified 24,000 duplication cases. The ordinary library containing the identical coding information ties on all 12,840 single-erasure cases at the new sizes. [31]

The preserved failures delimit the result. With two erasures, 1,560 of 6,600 five-event cases and 13,680 of 86,400 six-event cases remain ambiguous under parity. All 12,840 tested single-bit substitutions are detected, but a single parity check cannot generally correct an unknown flipped bit. Erasing parity while flipping AB can silently change ABCDE(F) into BACDE(F); flipping AB and AC can produce BCADE(F) with parity still satisfied. These cases lie outside the promised one-known-erasure model and are retained so that successors cannot quietly generalize the guarantee.

The new-size checks establish implementation consistency on fresh finite tasks. The general coding argument and the four-event design were already known before their execution.

Accordingly, this is technical transfer of a known procedure, not learned independent competence, prospective foresight or numerical retained Alpha.

6. A second construction: deterministic internal races, order-sensitive inputs

6.1 Exact hysteron model

The next search refinement sought a physically interpretable distinction: internal switching order should not alter the result, while the order of external events should. The constructed model contains binary states s_i, a held input vector h, and effective fields

$$H_i(s,h) = h_i + \sum_j C_{ij} s_j.$$

An inactive bit may switch on when $H_i \geq 1.00$; an active bit may switch off when $H_i \leq -0.80$. Every allowed single-bit update choice is explored. No update priority is used to manufacture determinism. Nominal calculations use exact integer-scaled values. [32]

For the four-bit state (x, y, z, w), only $C_{xz} = C_{zx} = 0.60$ and $C_{yw} = C_{wy} = 0.60$ are nonzero. Pulse A is $(1.20, 0, 0, 0.60)$, and pulse B is $(0, 1.20, 0.60, 0)$. Each is held until settling and followed by zero input. C holds A+B simultaneously; R applies $(-2, -2, -2, -2)$ until settling and then returns to zero. Two- and three-bit projections are evaluated as separate models. An independent Boolean recurrence supplies the semantic oracle:

$$x' = x \vee a, \quad y' = y \vee b, \quad z' = z \vee [b \wedge (x \vee a)], \quad w' = w \vee [a \wedge (y \vee b)].$$

R clears the state. Under the declared exactly-once two-event interpretation, AB produces 1110 and BA produces 1101. C, ABA and BAB all produce 1111. The collisions are central: the model cannot reconstruct full history or uniquely identify simultaneous events. Arbitrary finite-duration overlaps, thermal activation and physical switching dynamics are outside its settled-input abstraction. [32]

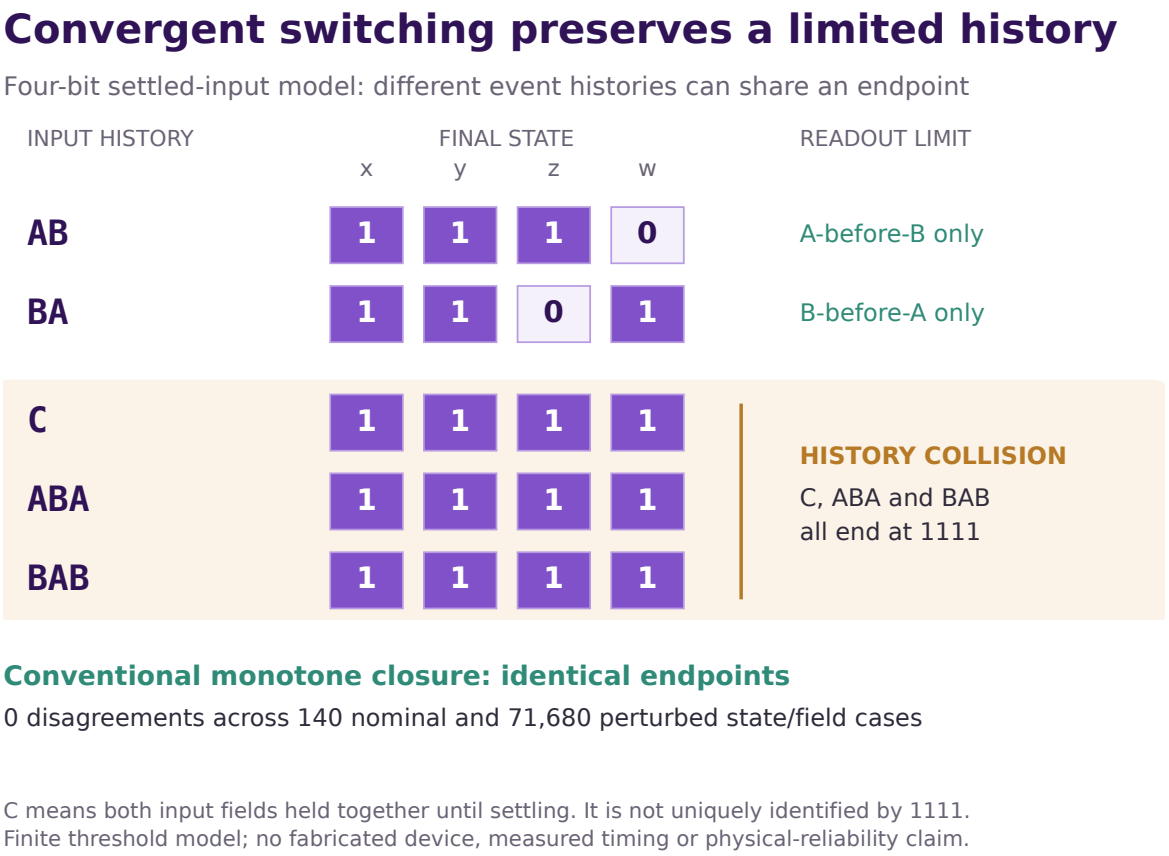


Figure 3. Four-bit abstract hysteron readouts distinguish B from B , while C, B and B B collide at 1111. Every allowed internal update schedule agrees within the declared model. conventional monotone-closure implementation gives identical results. This partial event-order construction does not satisfy the original four-event single-erasure offering.

6.2 Why it works, and why it does not earn novelty

Under A, B, C and zero, an active bit cannot switch off because fields are nonnegative and the off threshold is negative. Every on-switch can only increase neighboring fields. The process is therefore monotone on a finite Boolean lattice and reaches the same least fixed point under every maximal asynchronous update sequence. R forces all bits off. With symmetric couplings, the coarse energy

$$E(s;h) = -\sum_i h_i s_i - \sum_{i<j} C_{ij} s_i s_j$$

strictly decreases at every allowed fixed-input switch. This supports finite-model consistency; it supplies neither a continuous elastic energy landscape nor a manufactured device. Different external vector paths can still yield different final states. Internal confluence does not imply that AB and BA commute, and this result does not overturn scalar return-point-memory theory. [13, 32]

The confirmation includes all 140 nominal initial-state/held-field cases across two, three and four bits; 5,461 words of length zero through six per model over A/B/C/R; and 1,024 fresh seeded words of length one through 64 per model. There are no oracle disagree-

ments or nominal nonconfluent cases. Parameter checks use 512 draws per dimension, perturbing present symmetric edges, thresholds and nonzero drives by at most 0.04, giving 71,680 state/field checks without disagreement. Analytic interval margins also cover the continuous specified perturbation box; the smallest required gated-activation margin is 0.08. These are model checks, not independent physical trials or estimated real failure rates. [32]

Unfavorable controls remain executable. Reducing each coupling from 0.60 to 0.35 destroys order discrimination. An antagonistic pair terminates in either 01 or 10 depending on update order. An asymmetric feedback pair cycles. The checker reports these behaviors. Eight acceptance tests pass, including the negative controls.

The novelty verdict follows from stronger comparisons. Conventional monotone closure matches the proposed dynamics exactly. Mechanical counting, pattern recognition and reset already exist; systematic transition-graph synthesis already exists; vectorial sequential memory and autonomous event-order recorders are identified prior art. The autonomous recorder comparison is especially useful because its first-occurrence semantics are different from, and in some respects stronger than, this recurrent-precedence state. Direct full text was unavailable for the recent vectorial-memory preprint, so no detailed performance comparison to it is asserted. A working threshold motif earns a reference construction, not a new discovery claim. [9, 10, 13–15, 32]

7. Commercial delivery, pricing and financing for the same offering

7.1 Scope and calibrated assumptions

The commercial model concerns the original physically qualified four-event chronology device with one known erased readout. It does not sell the smaller latch, the abstract code or a generic diagnostic service. There is no equivalent human-service market price to halve. Instead, the comparison estimates a human-led development pathway and models buyer value separately. All buyers, volumes, physical success probabilities, device costs and prices are explicit scenarios; none is a quote, sale or demonstrated willingness to pay. [33]

Two external anchors calibrate development labor. Job Bank reports Montréal mechanical-engineer wages of CAD 30.43, 44.52 and 66.39 per hour, with a 2024 reference period and a 19 November 2025 update. The Bank of Canada’s retrieved 26 October 2026 indicative rate is CAD 1.4025 per USD. Loading assumptions of 1.45, 1.70 and 2.10 produce USD 31.46, 53.96 and 99.41 per engineering hour. The loading factors, hours and other costs are planning estimates, not source-reported project costs. [26, 27, 33]

Development estimate	Low	Central	High
Engineering hours	400	900	2,200
Physical test hours	200	500	1,500
External prototypes, tooling and tests	$18,000	$65,000	$220,000
Assumed duration	3 months	9 months	18 months
Human-led total H	$37,447	$141,170	$576,411
Expert AI, no labor benefit E0	$38,054	$141,792	$577,056

Physical test labor is costed at assumed USD 32, 45 and 65 per hour. Research financing uses 10% annually on an assumed average half-deployed balance over the stated duration. Fifteen percent of engineering cost represents failed screens already inside the total. The central E0 total consists of $48,567 engineering, $22,500 physical testing, $65,000 external work, $600 inference/tools and $5,125 financing. The $600 is a scenario allowance: 20 million input and four million output tokens at the documented `gpt-6-astra` rates yield $400, plus $200 tools and compute. It is not a provider invoice. [28, 33]

E0 assumes **no** AI reduction in engineering hours or increase in physical success. Hypothetical 25% and 50% engineering-hour reductions are sensitivities, with physical testing unchanged. A high human R&D estimate therefore cannot silently become evidence that an AI pathway is cheaper, faster or more likely to work.

7.2 Buyer participation and the best tested arrangement

For each simulated buyer, device value combines assumed avoided losses and staff savings, less false-alarm costs. The model assigns 70% prevented-loss benefit, but a passive record read after an event cannot undo its damage. A downstream decision—such as disposition, future prevention or adjudication—and its timing must be specified before this becomes calibrated value. A manual outside option receives its own assumed loss avoidance, staff saving and $3 per-unit cost. Incremental value is the positive difference between device value and the best modeled outside option. Buyers participate only when effective price plus the financing cost of any deposit is at most 70% of that incremental value. The remaining surplus supplies an explicit participation margin; the rule is an assumption, not an elicited demand curve or a truthful mechanism. [33]

The tested list-price grid is $2, $3, $4, $5, $6, $8, $10, $12, $16 and $20. Payment policies are net 30 or a 20% deposit with a 5% price discount. Deposit financing is charged to buyers at an assumed 10% annual rate for thirty days. The supplier’s credit-loss model distinguishes postpaid exposure. A lower effective price can increase participation while a deposit reduces financing pressure; raising prices never holds demand artificially fixed.

Development uses 128 worlds. Policies were saved before a fresh confirmation seed generated 512 worlds for each of three scenarios. Capacity is capped at 120,000 devices annually. Potential buyers number 18, 40 or 70, with central per-buyer volumes of 2,000, 3,000 or 3,500 respectively; reachability, buyer value, demand, volume and unit cost vary under the declared distributions. The selected policies are the best tested policies for this model and grid, not globally optimal real-world prices. [33]

Confirmation scenario	Selected list price and terms	Mean annual units if qualified	Mean annual operating profit if qualified	Expected project profit after central E0 development
Conservative	$4, net 30	2.9	−$73,910	−$160,270
Central	$5; 20% deposit, 5% discount	58,113	$20,979	−$131,303
Favorable	$8; 20% deposit, 5% discount	96,034	$417,868	$171,609

The assumed physical-qualification probabilities q are 0.25, 0.50 and 0.75 respectively. Expected project profit is $q\,\mathbb{E}[\Pi_{\text{operation}}]-A$, where A is the development total and the operating branch is entered only after qualification. The probabilities are not inferred from the code or hysteron checks. Conservative participation almost entirely fails; 511 of 512 worlds have no sales. In the central scenario, 196 of 512 qualified-operation worlds lose money. These failures stay in the evidence rather than disappearing behind an average. [33]

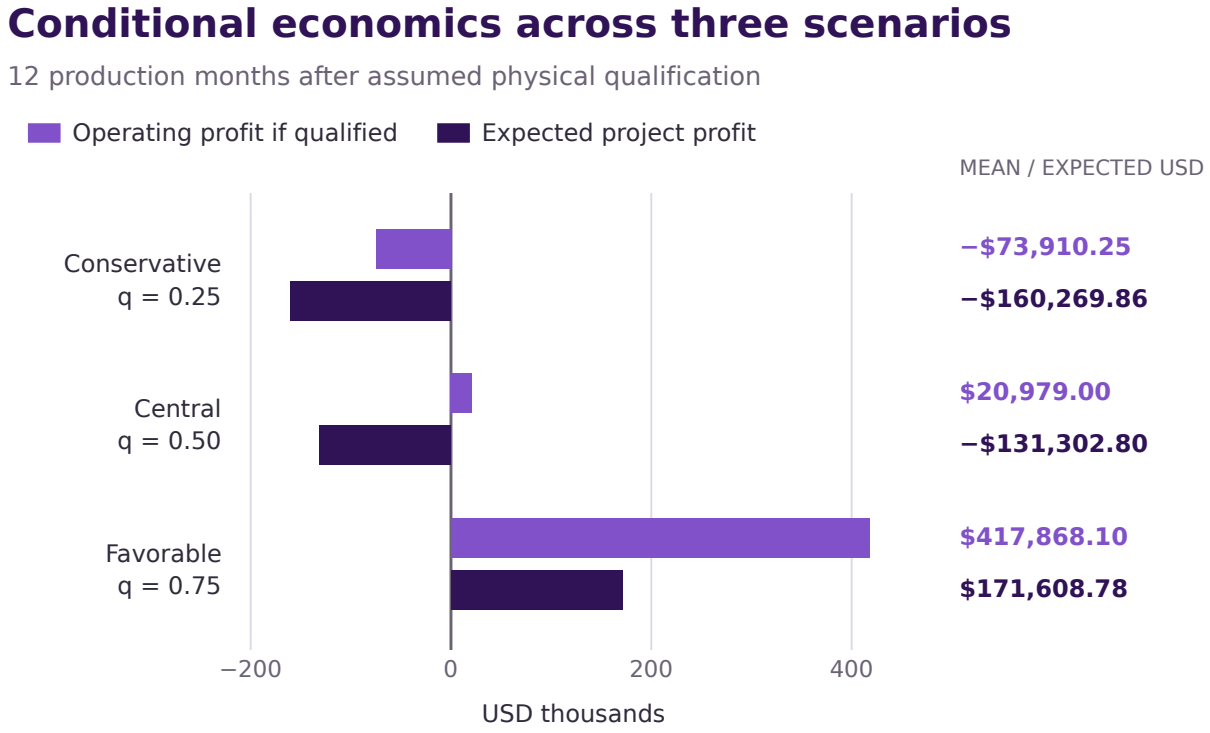


Figure 4. Frozen confirmation economics for the same hypothetical four-event chronology offering. Operating profit assumes a qualified product and twelve production months. Expected project profit is the stipulated physical-success probability times mean operating profit, minus central development cost; it is not an annualized return or a sampled project-world mean. Probabilities, customer value and cost adequacy remain unvalidated. No actual collections occurred.

7.3 Favorable-case accounts and unresolved cost scope

The favorable result is a pre-tax scenario calculation over development plus one operating year, not an after-tax return or net present value. Its 75% success assumption and central \$141,792 development estimate yield $0.75 \times 417,868 - 141,792 = 171,609$ expected project profit. Qualification failures and adverse commercial worlds still produce 137 losses among 512 simulated projects. The project-profit fifth and ninety-fifth percentiles are −\$141,792 and \$437,756. These are scenario quantiles, not empirical confidence intervals. [33]

Favorable qualified-operation account	Mean USD	Interpretation
Customer incremental value	\$1,326,650	Value above the modeled outside option, before price
Customer remaining surplus	\$595,594	After price and prepayment financing
Gross recognized revenue	\$729,856	96,034 units at effective price \$7.60, subject to rounding
Collected cash	\$724,017	After modeled credit losses and collection runoff
Production and delivery	\$153,144	Materials, fabrication, QA, packaging/shipping, support/recovery
Customer acquisition	\$18,117	Assumed acquisition expenditure
Warranty expense and credit loss	\$20,436	\$14,597 warranty plus \$5,839 credit loss
Contribution before overhead/finance	\$538,159	Revenue less the preceding variable/business charges
Fixed overhead and financing	\$120,290	\$120,000 overhead plus \$290 operating finance
Annual operating profit	\$417,868	Before the separately reported originating development cost
Warranty liquidity reserve	\$21,896	Reserved cash, not a second warranty expense
Available after reserve and debt	\$395,972	Conditional operating cash; not actual growth funds

Unit production cost is decomposed into materials 40%, fabrication 25%, quality assurance 15%, packaging/shipping 12%, and support/recovery 8%. These are broad allowances, not an evidenced complete bill of materials. Failed-prospect effort, yield/rework, equipment, qualification, rights, insurance and exceptional returns are unscoped. The 8% support/recovery allocation may overlap the separate warranty charge; internal physical-test labor and external testing also lack task-level separation. Arithmetic consistency cannot establish that real costs are complete or counted once. Fixed overhead and acquisition are separate model entries. The twelve-month delivery model includes a thirteenth month for receivable and warranty runoff. Deposits, production and assumed acceptance occur sequentially within a monthly bucket. The thirty-day lead is a buyer opportunity-cost assumption, not dated seller cash evidence. Deposits become revenue only at assumed acceptance; credit principal is financing, not revenue. Warranty expense is 2% of revenue,

while a distinct 3% liquidity reserve affects available cash without being expensed twice. [33]

A revolving facility at an assumed 10% annual rate covers operating cash deficits. In the favorable case, mean peak credit is $31,756. Adding central development gives mean capital required of $173,548, with a ninety-fifth percentile of $179,382. These are period-end requirements under coarse monthly timing, not committed capital: intramonth peaks, procurement leads, sales ramps, refunds and delayed acceptance are unresolved. Actual collections, deliveries, recognized revenue and token receipts are all zero. The actual full attributable session cost is unknown, so no actual profit or available growth fund is asserted. [33, 34]

7.4 Robustness, losses and the development decision

At the favorable base assumptions, the break-even qualification probability is approximately 0.339. This threshold holds the modeled operating distribution and central development cost fixed. Central operating economics would require $q > 1$ to repay that development cost within the modeled horizon. Higher R&D costs can overturn the favorable case: the high E0 development estimate produces approximately −$263,655 expected project profit; even an assumed 50% engineering-hour reduction leaves it negative. [33]

An exploratory demand/cost grid keeps the selected policies fixed. Twenty-four of 42 favorable grid cells have both positive expected project profit and a positive fifth-percentile annual operating profit; no central or conservative cell does. At half base demand and unchanged unit cost, favorable annual operating profit remains approximately $160,565, yet expected project profit becomes −$21,368. At 75% demand and 1.5 times unit cost, expected project profit is about $41,179. These bounds clarify which economic assumptions deserve early calibration. They are not a demonstrated robust market region. [33]

A same-team independent Decimal audit checked 16,968 identities or comparisons across all 1,536 confirmation worlds; the saved reconstructed auditor adds 12,543 numeric checks. Residuals remain below $0.000001. The original inline command was not preserved; its reconstructed source and changed rounding residual are explicitly documented. This validates arithmetic, failure preservation and physical-work invariance, not actionable benefit, real cost completeness or disjoint cost scope. The model-review record preserves these material limitations without changing the frozen results. [33] All actual decisions remain **HOLD: unqualified physical offering and failed novelty gate**. Avoiding future expenditure does not erase sunk research cost.

7.5 Post-result cost headroom

An additional exploratory accounting stress, performed after the frozen results, asks how much unscoped cost the favorable case could absorb without changing buyer behavior, prices, qualification or financing. Let ΔA be additional originating cost and ΔO additional annual qualified-operation expense. Positive expected project profit requires $\Delta A + 0.75\Delta O < 171{,}608.78$ USD. Keeping its fifth-percentile annual operating result positive additionally requires $\Delta O < 191{,}951.16$ USD for a constant annual expense. The central case has no nonnegative added-cost region. These are accounting boundaries inside the constructed scenario, not estimates of omitted costs, evidence of cost completeness or a behavioral reoptimization. Induced financing, delays and demand responses require a new model. The exact derivation and provenance are in results/cost-headroom-exploratory.json and src/cost_headroom.py. Actual HOLD is unchanged. [33]

8. Frontier models, Operational Alpha and the foresight comparison

Official documentation was checked on 27 October 2026. The inventory records exact identifiers, rates, proposed configurations and access limitations. It establishes suitable experiment candidates, not that the present account had every listed API entitlement or that provider benchmarks predict success on this offering. No paid provider API mission experiment was run. [28–30]

Exact model identifier	Proposed role	Proposed reasoning/output configuration	USD per million input/output tokens
`gpt-6-astra`	Discovery and difficult judgment	`max`, 64,000 output cap	10 / 50
`gpt-6.1-sol`	Execution comparator	`high`, 32,000 output cap	2 / 10
`gpt-6.1-terra`	Computer-use execution candidate	`high`, 32,000 output cap	1 / 5
`claude-fable-5-1`	Independent-family validation candidate	Adaptive thinking, `max`, 64,000 cap	10 / 50
`claude-opus-5-5`	Implementation comparator	Adaptive thinking, `high`, 64,000 cap	4 / 20
`gemini-4-argon`	Scientific reasoning candidate	`high`, 65,536 output cap	2 / 10 through 30 November 2026

The inventory records context limits, selected provider tool availability and the Google rate change scheduled for December. The listed configurations are proposals, not executed mission configurations; exact tool schema versions must be frozen before a comparative provider run. OpenAI reasoning/tool proposals use the Responses API and omit unsupported sampling controls. Actual cost logs must include reasoning output, retries, verification and tool usage. Closed weights, unrestricted consumer-subscription automation and untested account access are not assumed.

Operational Alpha means E performing better than U with the **same** frontier model, initial evidence, tools, reasoning limits and total budget. Retained Alpha compares E+ with E after charging the incremental retained capability. The present study implements expert research practices—source construction, decomposition, exact verification, adversarial controls and recovery—but contains no controlled numerical E-versus-U result. H is a transparent development estimate, not a recruited human team. Independent implementations were produced within the same assistant-controlled team; they are not independent model families or blinded human adjudicators. [31–35]

A prospective comparison should therefore be frozen before its next discovery run. The proposed design uses identical need/resource packets containing no intended opportunity, source access cutoffs, equal model identifiers and monetary/token limits, and a fixed candidate budget. Representative U receives a competent open-ended opportunity-search instruction and the same available tools; E receives the expert orchestration procedure. Expertise may change decomposition and verification, but cannot obtain extra information or budget. Multiple sessions and seed assignments should be specified in the frozen protocol, rather than selected after favorable outputs appear.

Novelty, usefulness, technical feasibility and commercial plausibility should be reviewed separately against predeclared criteria. Reviewers should receive source-backed constructions without condition labels where practical. Human experts should search the same packet if recruited; otherwise the edition must report that the human comparison is absent or explicitly modeled. The final test must include an executable construction or decisive experiment, with fresh acceptance cases hidden from the search sessions. Merely failing to mention an opportunity in a baseline sample does not establish beyond-human foresight. Historical replay would additionally require defensible information cutoffs and leakage controls; none is claimed here.

9. Insight, Nova-Seed, FusionPlan and MARK as an actual connected mechanism

The integration copies and hashes forty unchanged adapter source/license files and two additional unchanged scenario modules from the audited repository snapshot. It executes native scenario analysis, Nova-Seed cryptographic roundtripping, FusionPlan compilation and verification, cost-ledger safeguards, and authority checks. This is a real local integration with public fixtures, distinct from a narrative business plan. [34, 35]

The native Insight scenario receives seven reviewer-supplied rejected candidates and retrieves eight exact excerpts from their supplied evidence. Each candidate has unit synthetic cost and zero eligibility value. Across 128 possible subsets it selects none, produces no schedule, and `makeGenome` refuses to create a venture genome. Identical numeric coordinates place all seven candidates on the computed nondominated frontier, which does not make them qualified. The numeric setup records the completed review; it is not an independent semantic novelty evaluation or proof that Insight discovered the rejection unaided. The generic refusal suggestion to increase budget does not override the actual reason: eligibility is zero despite every candidate fitting the synthetic budget.

A separate research Nova-Seed preserves the negative result, opportunity, original purpose, sources, uncertainty, rights scope, transformation witness, evidence and next experiment. Native WebCrypto encryption and authenticated decryption roundtrip the object; wrong passphrases and modified commitments are rejected. A public fixture passphrase establishes an inspectable test, not secrecy. Cryptographic integrity shows that the object is unchanged, not that its scientific claim is true.

The FusionPlan contains four executable milestone descriptions: discovery, construction, economics and retention. Native Merkle compilation and verification check committed fields. The fixture allocates 100 base units of AGIALPHA per job, 400 total; these are local test units, not USD, funded bounties or revenue. Dependencies, full budgets, permissions, acceptance conditions and falsifiers are retained in the research object. The native commitment format does not itself enforce every wider research obligation.

Thirteen adapter checks pass. They include rejecting a substituted acceptance metric, double-charged activities, treating unknown cost as zero, unsupported capsule scope, unreviewed rights and production admission based only on a valid local signature. A simulated identity without live ENS authorization remains denied, and denial does not consume the challenge. The public research artifact cannot be renamed to fit an existing production capability’s narrower approved scope. The adapter emits an inspectable MARK review packet; MARK execution remains `NOT_EXECUTED`, and no minting, listing, trading, funds movement, on-chain action or expanded permission occurs. [34]

These outcomes preserve the existing validator roles. The research pipeline can archive

a useful failed hypothesis while venture promotion and production access remain closed. Native execution used Python 3.12.14 and Node 24.22.1; the production Node pin differs, so the local checks are not a production release qualification.

10. What the successor inherits—and what it must earn

The successor can inherit exact models, source-scoped claims, counterexamples, configurations, a schedule-complete checker, an independent decoding oracle and the complete commercial assumptions. Each object has a version, provenance, dependencies, scope and failure modes. The intended rights scope is public research code and public-source metadata; external publications retain their own rights. Secrets, customer material and protected evaluation answers are not part of the capsule. [31–35]

Accounting separates A, originating delivery or research; B, additional evaluation, extraction, retention, invocation and maintenance; and C, optional extra research or Gym training. Necessary learning already charged to A is not charged again in B. The present construction was unpaid research, not learning acquired during an accepted paid job. The actual native ledger marks originating and additional-validation expenditure unknown, sets optional Gym execution to zero, and refuses to calculate actual profit from incomplete costs.

The optional retained-service scenario adds $2,500 extraction/validation and $1,200 annual maintenance, or $3,700. Ten future missions would require an average $370 incremental benefit per mission to break even. The identical-information ordinary library produces no observed acceptance improvement over the retained code. Under the tested zero-incremental-saving case, the optional premium retention investment loses $3,700 and is rejected. Public preservation remains useful; it is not evidence that maintaining a specialist business is free or profitable. [33]

No model weights were trained, no Gym learning curve was measured and no specialist superhuman claim is made. The fresh five- and six-event tasks demonstrate the same known algorithm operating at new sizes. Remote service access would likewise not establish that a recipient independently acquired competence. The operational inheritance rule is: **REUSE stable structure; RECHECK the new mission’s meaning; REGRANT authority separately.** Authenticated agent-network distribution must use the actual authorization workflow; local fixture identities do not grant access across `*.agent.agi.eth`.

The next challenge is a useful physical history property that existing mechanical, finite-state and coding controls cannot satisfy under the same complete resource budget. The inherited models help falsify proposals; they do not establish that such a gap exists.

11. The smallest useful next validation

The first next step is a bounded novelty-and-realizability challenge, before a full hardware program. Specify one valuable history predicate, its event and fault model, complete physical component budget, permissible energy and readout mechanism. Ask a materials specialist to compare it with the identified general hysteron-synthesis, event-recorder and classical-code controls. A proposed advantage fails if a conventional construction meets the same obligations at comparable total resources.

If a functional gap survives, map the smallest motif to a calibrated continuous potential or existing tunable-hysteron geometry. Check metastable branches, actual ramps, damping, noise and reset; then fabricate only the component needed to falsify the proposed physical mechanism. This step would test physical feasibility. Fabricating the already known par-

ity or latch principle alone would not repair the novelty deficit. The complete four-event erasure requirement remains the product acceptance gate.

In parallel, a small scoped buyer interview can test whether the claimed history changes a real decision, what outside option is used, and whether the favorable model's incremental value and volume assumptions are plausible. A quote can calibrate fabrication and test costs. These activities refine a conditional model; they do not retroactively turn simulated revenue into collections. Their budgets and permissions must be separately authorized when needed.

The exact constructions, conventional ties and rejected opportunities equip a successor to continue the search with its acceptance standard intact.

Appendix A. Reproducibility and evidence map

All paths below are relative to this study directory. The repository edition contains source, frozen configurations, seeds, results, audit records and figures. The core order-memory and confluence experiments use Python's standard library; figure reproduction additionally uses matplotlib. The integration requirements and exact upstream source lock are separate from the scientific kernels.

Evidence	Reproduction entry point	Preserved output
Four-event probe	discovery/order-memory/mechanical_memory_probe.py	`mechanical_memory_probe.json`
Frozen new-size decoding	discovery/order-memory/order_transfer_study.py	transfer_validation/transfer_results.json, compressed case stream
Hysteron confirmation	experiments/confluence/study.py; `test_study.py`	`results.json`, `freeze.json`, confirmation words
Economics	src/economics.py with config/economics.json	Development, policy freeze, confirmation worlds and sensitivity
Native integration	integration/run_adapter.py; `run_scenario.mjs`	Chronology and scenario run reports

Consult each component's README or reproduction notes for its working directory and arguments. Frozen configurations must be preserved when reproducing results; changing a model requires a new labeled experiment, not overwriting the acceptance history. The order study freezes source/configuration before n = 5/6 execution. Confluence freezes its configuration before programmatic confirmation, after the design was hand-derived. Economics freezes selected policies before fresh-seed confirmation. These are distinct chronological guarantees and none makes the overall adaptive search blind.

The economics development seed is `2026102701`; confirmation uses `2026102702`. All 1,536 confirmation worlds are retained. An interrupted disk-write artifact and its unchanged rerun are preserved. Independent audit files document decoding and Decimal accounting checks. Figures are generated by src/build_figures.py; figures/caption_manifest.json records their source data and scope. Deterministic exhaustive counts need no sampling confidence interval; commercial and perturbation scenarios do not become empirical reliability estimates merely because many cases were run.

Appendix B. Contributions, data availability and publication status

The author line follows the commissioned manuscript. The user supplied the mission, novelty gate, research obligations and ecosystem references. Assistant-controlled research agents supplied candidate generation, source screening, mathematical formulations, code, controls, economic scenarios, integration adapters and drafting. Separate agents performed implementation audits, but shared the same controller and evidence environment. No independent human expert study, external peer review, live buyer trial or provider-controlled model comparison is implied.

Data and code are supplied with this study, including failed cases and the no-flagship decision. Scientific acceptance applies only to the declared finite computations. Commercial figures are conditional simulations. Production authorization remains denied. This manuscript is a source draft for review; the canonical browser/PDF publication route is coordinated with the existing owner. No merge, live canonical Pages update or external publication is asserted by this document.

References and source scope

All online sources below were inspected or their specified metadata checked during the 27 October 2026 study. Local evidence references identify reproducible artifacts. Descriptions are scoped to the claims used here.

1. **Commissioned brief.** Exact supplied instructions, discovery/INITIAL_BRIEF.txt. Primary task record, 2026-10-27.
2. **MONTREAL.AI.** AGI-Alpha-Agent-v0 and coordination issue 4850. Primary implementation and coordination records; inspected snapshot recorded in provenance.
3. **Transformational Intelligence.** Foundational paper. Primary project research, inspected 2026-10-27.
4. **Invent the Economy.** Foundational paper. Primary project research, inspected 2026-10-27.
5. **Autonomous Enterprises: Build the Means of Building.** Foundational paper. Primary project research, inspected 2026-10-27.
6. **MONTREAL.AI commercial and retained-capability studies.** Repository primary papers: Complete Economics, Profitable Work First, When Paid Work Teaches the Network, and The Network Learns to Invent. Snapshot scope and corrections in the audit; no results imported.
7. **Operational Alpha v2.** Browser edition at inspected commit. Primary project study, 2026; local inspected record in provenance.
8. **Emergent disorder and mechanical memory in periodic metamaterials.** Nature Communications, 2024. Primary journal article; open full text.
9. **Liu et al. Controlled pathways and sequential information processing in serially coupled mechanical hysterons.** PNAS, 2024. Primary journal article; open full text.
10. **Teunisse and van Hecke. Transition graphs of interacting hysterons: structure, design, organization and statistics.** Royal Society Open Science, 2025-09-24. Primary full text: PMC.
11. **Paulsen. Mechanical hysterons with tunable interactions of general sign.** Nature Communications, 2026-04-11. Primary journal article.
12. **Hamming. Error Detecting and Error Correcting Codes.** Bell System Technical Journal record, 1950. Primary foundational paper; PDF copy.
13. **Sethna et al. Hysteresis and hierarchies: Dynamics of disorder-driven first-order phase transformations.** Physical Review Letters, 1993; primary preprint.
14. **Meulblok and van Hecke.** Vectorial-input mechanical memory preprint, July 2026. Primary abstract inspected; direct full text unavailable during the probe.
15. **Cardelli.** Autonomous event-order recording, 2023. Primary research chapter, full text inspected; first-occurrence semantics differ from this study's recurrent-precedence latch.

16. **Shetty and Braga-Neto. LLM-IDEA: Identifiability-Driven Experimental Agent for Autonomous Discovery of Mechanistic World Models.** Primary preprint, 2026-10-08.
17. **Rassouli, Rosas and Gunduz. Data Disclosure under Perfect Sample Privacy.** Primary preprint, 2019-04-02. Its distributional privacy notion is not unrestricted side-information robustness.
18. **Zhao et al. From Mathematical to Executable Certificates for Machine Unlearning.** Primary preprint record, 2026-10-01. Metadata inspected; detailed technical performance not used.
19. **Hou et al. Experimental realization of a full-band wave antireflection based on temporal taper metamaterials.** Communications Physics, 2026-01-15. Primary full text.
20. **Catalytic resonance theory: parallel reaction pathway control.** Chemical Science, 2020. Primary article; open full text.
21. **Rajput and Pahlavan. Fluidic Hysterons and Memory in Flow Networks.** Physical Review X, 2026-10-09. Primary abstract inspected.
22. **Ahrar, Raje, Lee and Hui. Pneumatic computers for embedded control of microfluidics.** Science Advances full text, 2023. DOI: 10.1126/sciadv.adg0201.
23. **Generalizing multiple memories from a single drive: The hysteron latch.** Primary full text, 2025.
24. **Ali et al. Materializing split, mixed, and three-body interactions using rotor-based mechanical hysterons.** Primary preprint, 2026-07-24. Abstract inspected.
25. **Thakkar, Rivas-Padilla and Arrieta. Visco-Bits: temporal-coding-inspired viscoelastic metamaterial for mechanical memory expansion.** Primary journal record, 2026; metadata. Year used because online and issue dates differ.
26. **Government of Canada, Job Bank. Mechanical engineer wages, Montréal.** Official wage report. Reference period 2024; updated 2025-11-19; retrieved 2026-10-27.
27. **Bank of Canada. Daily exchange rates.** Official indicative rates. USD observation for 2026-10-26, retrieved 2026-10-27; not a transaction quote.
28. **OpenAI.** Official models, latest-model guide and pricing. Retrieved 2026-10-27; exact inventory in provenance/models_2026-10-27.json.
29. **Anthropic.** Official Fable 5.1, Opus 5.5 and effort configuration. Retrieved 2026-10-27; access not tested.
30. **Google.** Official Gemini 4 Argon, latest-model documentation and pricing. Retrieved 2026-10-27; time-limited price recorded.
31. **This study: discovery and order-memory evidence.** discovery/order-memory/discovery_record.json, `mechanical_memory_probe.json`, transfer_validation/transfer_results.json, compressed cases, theory notes and independent adversarial review. Primary executable research, 2026-10-27.
32. **This study: confluence evidence.** experiments/confluence/README.md, model and configuration, freeze, source, tests and `results.json`. Primary executable research, 2026-10-27.
33. **This study: commercial evidence.** config/economics.json, src/economics.py, results/economics-results.json, frozen policy, all worlds, sensitivity, independent Decimal audits and results/economics-model-review-v1.json. Primary conditional simulation, 2026-10-27.
34. **This study: native workflow evidence.** integration/README.md, `SCENARIO_README.md`, source lock and reports under integration/results/. Primary local execution with fixture identities, 2026-10-27.
35. **This study: ecosystem audit.** provenance/AUDIT.md, `source_manifest.json` and inspected Operational Alpha record. Primary provenance and scope audit, 2026-10-27.